Windows Unified Host Utilities

SAN hosts and cloud clients

NetApp
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Windows Unified Host Utilities

Install Windows Unified Host Utilities 7.2

The Windows Unified Host Utilities (WUHU) enable you to connect a Windows host computer to NetApp storage systems.

The Windows Unified Host Utilities supports the following versions of Windows:

- Windows 2022
- Windows 2019
- Windows 2016
- Windows 2012R2
- Windows 2012

Windows Unified Host Utilities includes an installation program that sets the required Windows registry and Host Bus Adapter (HBA) parameters so that a Windows host can correctly handle the storage system behaviors for NetApp ONTAP and E-Series platforms.

When you install the Host Utilities software, the installer sets the required Windows registry and HBA parameters.

The following programs and files are installed on the Windows host computer. The default directory is `C:\Program Files\NetApp\Windows Host Utilities`.

Program	Purpose
`emulexhba.reg`	Troubleshooting program; run this program only if instructed to do so by technical support personnel.
`\NetAppQCLI\fcconfig.exe`	Used by the installation program to set the HBA parameters.
`\NetAppQCLI\fcconfig.ini`	Used by the installation program to set the HBA parameters.
`\NetAppQCLI*.*`	Used by the installation program to set the QLogic FC HBA parameters.
`san_version.exe`	Displays the version of the Host Utilities and FC HBAs.

The host utilities support different Windows host configurations, protocols, and multipathing options. For more information, see the NetApp Interoperability Matrix Tool.

Verify your host and storage system configuration

Before you install the host utilities, you must verify that the host utilities version supports your host and storage system configuration so that the software installs correctly.

Steps

1. Check the supported configuration in the NetApp Interoperability Matrix Tool.

2. Check the hotfixes required for the respective host in the SAN host Windows documentation.

> The Using Windows server 2022 with ONTAP document provides instructions on installing Windows hotfixes for Windows server 2022. Refer to the Windows documents in the host configurations category to find the relevant hotfix information for earlier versions of Windows server.

3. Add the iSCSI, FCP, or NVMe-oF license and start the target service.

> The FC and iSCSI protocols do not require licenses on E-Series storage systems using the SANtricity Storage Manager.

4. Verify your cabling.

   See the SAN configuration reference documentation for your version of ONTAP or E-Series hardware cabling for detailed cabling and configuration information.

Configure FC HBAs and switches

Install and configure one or more supported FC host bus adapters (HBAs) for FC connections to the storage system.

The Windows Host Utilities installer sets the required FC HBA settings.

> Do not change the HBA settings manually.

Steps

1. Install one or more supported FC HBAs according to the instructions provided by the HBA vendor.
2. Obtain the supported HBA drivers and management utilities and install them according to the instructions provided by the HBA vendor.
3. Connect the HBAs to your FC switches or directly to the storage system.
4. Create zones on the FC switch according to your FC switch documentation.
5. For ONTAP, zone the switch by the WWPN. Be sure to use the WWPN of the logical interfaces (LIFs) and not the WWPN of the physical ports on the storage controllers. See the SAN configuration reference documentation for more information.

Install the Host Utilities

The installation program installs the host utilities package and sets the Windows registry and HBA settings.

You must specify whether to include multipathing support when you install the Windows Unified Host Utilities software package. The installer prompts you for the following options. You can also run a quiet (unattended) installation from a Windows command prompt.

Multipathing support

- Choose `MPIO` if you have more than one path from the Windows host or virtual machine to the storage system.
- Choose `no MPIO` only if you are using a single path to the storage system.

The MPIO selection is not available for Windows XP and Windows Vista systems; multipath I/O is not supported on these guest operating systems.
For Hyper-V guests, raw (pass-through) disks do not appear in the guest OS if you choose multipathing support. You can either use raw disks, or you can use MPIO, but you cannot use both in the guest OS.

You can install the host utilities interactively or using the command line. The new Host Utilities installation package must be in a path that is accessible by the Windows host. Follow the instructions for installing the Host Utilities interactively or from the Windows command line.

Install interactively

To install the Host Utilities software package interactively, you must run the host utilities installation program and follow the prompts.

Steps

1. Download the executable file from the NetApp Support Site.
2. Change to the directory where you downloaded the executable file.
3. Run the `netapp_windows_host_utilities_7.2_x64` file and follow the instructions on the screen.
4. Reboot the Windows host when prompted.

Install from a command line

You can perform a quiet (unattended) installation of the host utilities by entering the appropriate commands at the Windows command prompt. The system automatically reboots when the installation is complete.

Steps

1. Enter the following command at the Windows command prompt:

```
msiexec /i installer.msi /quiet MULTIPATHING= {0 | 1}
[INSTALLDIR=inst_path]
```

   - `installer` is the name of the `.msi` file for your CPU architecture.
   - MULTIPATHING specifies whether MPIO support is installed. The allowed values are "0" for no and "1" for yes.
   - `inst_path` is the path where the host utilities files are installed. The default path is `C:\Program Files\NetApp\Windows Host Utilities\`.

> To see the standard Microsoft Installer (MSI) options for logging and other functions, enter `msiexec /help` at the Windows command prompt. For example, the `msiexec /i install.msi /quiet /l*v <install.log> LOGVERBOSE=1` command displays logging information.

Install Windows Unified Host Utilities 7.1

The Windows Unified Host Utilities (WUHU) enable you to connect a Windows host computer to NetApp storage systems.

The Windows Unified Host Utilities supports the following versions of Windows:

- Windows 2022
- Windows 2019
- Windows 2016
- Windows 2012R2
- Windows 2012

Windows Unified Host Utilities includes an installation program that sets the required Windows registry and Host Bus Adapter (HBA) parameters so that a Windows host can correctly handle the storage system behaviors for NetApp ONTAP and E-Series platforms.

When you install the Host Utilities software, the installer sets the required Windows registry and Host Bus Adapter (HBA) parameters.

The following programs and files are installed on the Windows host computer. The default directory is `C:\Program Files\NetApp\Windows Host Utilities`.

Program	Purpose
`emulexhba.reg`	Troubleshooting program; run this program only if instructed to do so by technical support personnel.
`\NetAppQCLI\fcconfig.exe`	Used by the installation program to set HBA parameters.
`\NetAppQCLI\fcconfig.ini`	Used by the installation program to set HBA parameters.
`\NetAppQCLI*.*`	Used by the installation program to set QLogic FC HBA parameters.
`san_version.exe`	Displays the version of the Host Utilities and FC HBAs.

The Host Utilities support different Windows host configurations, protocols, and multipathing options. See the NetApp Interoperability Matrix Tool for the most current list of supported configurations.

Verify your host and storage system configurations

Before you install the Host Utilities, you must verify that the Host Utilities version supports your host and storage system configuration so that the software installs correctly.

Steps

1. Check the supported configuration in the NetApp Interoperability Matrix Tool.
2. Check the hotfixes required for the respective host on the SAN host Windows documentation.

> The Using Windows server 2022 with ONTAP document provides instructions on installing Windows hotfixes for Windows server 2022. Refer to the Windows documents in the host configurations category to find the relevant hotfix information for earlier versions of Windows server.

3. Add the iSCSI or FCP license and start the target service.

> The FC and iSCSI protocols do not require licenses on E-Series storage systems using the SANtricity Storage Manager.

4. Verify your cabling

   See the SAN configuration reference documentation for your version of ONTAP or E-Series hardware cabling for detailed cabling and configuration information.

Configure FC HBAs and switches

Install and configure one or more supported FC host bus adapters (HBAs) for FC connections to a storage system.

The Windows Host Utilities installer sets the required FC HBA settings.

> Do not change the HBA settings manually.

Steps

1. Install one or more supported FC HBAs according to the instructions provided by the HBA vendor.
2. Obtain the supported HBA drivers and management utilities and install them according to the instructions provided by the HBA vendor.
3. Connect the HBAs to your FC switches or directly to the storage system.
4. Create zones on the FC switch according to your FC switch documentation.
5. For ONTAP, zone the switch by worldwide port name (WWPN). Be sure to use the WWPN of the LIFs and not of the physical ports on the storage controllers. See the SAN configuration reference documentation for more information.

Install the Host Utilities

The installation program installs the Host Utilities package and sets the Windows registry and HBA settings.

You must specify whether to include multipathing support when you install the Windows Unified Host Utilities software package. The installer prompts you for the following choice. You can also run a quiet (unattended) installation from a Windows command prompt.

Multipathing support

- Choose `MPIO` if you have more than one path from the Windows host or virtual machine to the storage system.
- Choose `no MPIO` only if you are using a single path to the storage system.

The MPIO selection is not available for Windows XP and Windows Vista systems; multipath I/O is not supported on these guest operating systems.
For Hyper-V guests, raw (pass-through) disks do not appear in the guest OS if you choose multipathing support. You can either use raw disks, or you can use MPIO, but you cannot use both in the guest OS.

You can install the host utilities interactively or using the command line. The new Host Utilities installation package must be in a path that is accessible by the Windows host. Follow the instructions for installing the Host Utilities interactively or from the Windows command line.

Install interactively

Steps

To install the Host Utilities software package interactively, you must run the Host Utilities installation program and follow the prompts.

Steps

1. Download the executable file from the NetApp Support Site.
2. Change to the directory from which you downloaded the executable file.
3. Run the `netapp_windows_host_utilities_7.1_x64` file and follow the instructions on the screen.
4. Reboot the Windows host when prompted.

Install from a command line

You can perform a quiet (unattended) installation of the Host Utilities by entering the appropriate commands at a Windows command prompt. The system automatically reboots when the installation is complete.

Steps

1. Enter the following command at a Windows command prompt:

```
msiexec /i installer.msi /quiet MULTIPATHING= {0 | 1}
[INSTALLDIR=inst_path]
```

   - `installer` is the name of the `.msi` file for your CPU architecture
   - MULTIPATHING specifies whether MPIO support is installed. Allowed values are "0" for no, "1" for yes
   - `inst_path` is the path where the Host Utilities files are installed. The default path is `C:\Program Files\NetApp\Windows Host Utilities\`.

> To see the standard Microsoft Installer (MSI) options for logging and other functions, enter `msiexec /help` at a Windows command prompt. For example, the`msiexec /i install.msi /quiet /l*v <install.log> LOGVERBOSE=1` command displays logging information.

Upgrade the Windows Unified Host Utilities

The new Host Utilities installation package must be in a path that is accessible by the Windows host. Follow the instructions for installing the Host Utilities interactively or from the Windows command line in order to upgrade the installation package.

Upgrade interactively

To upgrade the Host Utilities software package interactively, you must run the Host Utilities installation program and follow the prompts.

Steps

1. Change to the directory where you downloaded the executable file.
2. Run the executable file and follow the instructions on the screen.
3. Reboot the Windows host when prompted.
4. After the reboot completes, check the host utility version:
   a. Open **Control Panel**.
   b. Go to **Program and features** and check the host utility version.

Upgrade from a command line

You can perform a quiet (unattended) upgrade of the new Host Utilities by entering the appropriate commands at the Windows command prompt.

Steps

1. Enter the following command at the Windows command prompt:

```
msiexec /i installer.msi /quiet MULTIPATHING= {0 | 1}
[INSTALLDIR=inst_path]
```

   - `installer` is the name of the `.msi` file for your CPU architecture.
   - MULTIPATHING specifies whether MPIO support is installed. The allowed values are "0" for no and "1" for yes.
   - `inst_path` is the path where the Host Utilities files are installed. The default path is `C:\Program Files\NetApp\Windows Host Utilities\`.

> To see the standard Microsoft Installer (MSI) options for logging and other functions, enter `msiexec /help` at the Windows command prompt. For example, the `msiexec /i install.msi /quiet /l*v <install.log> LOGVERBOSE=1` command displays logging information.

The system automatically reboots when the installation is complete.

Repair and remove the Windows Unified Host Utilities

You can use the **Repair** option of the Host Utilities installation program to update the Host bus adapter (HBA) and Windows registry settings. You can also remove the Host Utilities entirely, either interactively or from the Windows command line.

Repair or remove interactively

The **Repair** option updates the Windows registry and FC HBAs with the required settings. You can also remove the Host Utilities entirely.

Steps

1. Open Windows **Programs and Features** (Windows Server 2012 R2, Windows Server 2016, Windows Server 2019, and Windows 2022).
2. Select **NetApp Windows Unified Host Utilities**.
3. Select **Change**.
4. Select **Repair** or **Remove**, as needed.
5. Follow the instructions on the screen.

Repair or remove from command line

The **Repair** option updates the Windows registry and FC HBAs with the required settings. You can also remove the Host Utilities entirely from a Windows command line.

Steps

1. Enter the following command on the Windows command line to repair Windows Host Utilities:

   ```
   msiexec /f installer.msi [/quiet]
   ```

   - `/f` repairs the installation.
   - `installer.msi` is the name of the Windows Host Utilities installation program on your system.
   - `/quiet` suppresses all feedback and reboots the system automatically without prompting when the command completes.

Configure registry settings

The Host Utilities require certain registry and parameter settings to verify that the Windows host correctly handles the storage system behavior.

Windows Host Utilities set the parameters that affect how the Windows host responds to a delay or loss of data. The particular values have been selected to verify that the Windows host correctly handles events such as the failover of one controller in the storage system to its partner controller.

Not all values apply to the device-specific module (DSM) for SANtricity Storage Manager; however, any overlap of values set by the Host Utilities and those set by the DSM for SANtricity Storage Manager do not result in conflicts.

FC, NVMe/FC, and iSCSI HBAs also have parameters that you must set to ensure best performance and to successfully handle storage system events.

The installation program supplied with Windows Unified Host Utilities sets the Windows, FC, and NVMe/FC HBA parameters to the supported values.

You must manually set the iSCSI HBA parameters.

The installer sets different values depending on whether you specify multipath I/O (MPIO) support when

running the installation program.

> You should not change these values unless NetApp technical support directs you to do so.

Registry values set by Windows Unified Host Utilities 7.2

The Windows Unified Host Utilities installer automatically sets registry values that are based on the choices that you make during the installation. You should be aware of these registry values and the operating system version.

The following values are set by the Windows Unified Host Utilities installer. All values are in decimal unless otherwise stated.

> HKLM is the abbreviation for `HKEY_LOCAL_MACHINE`.

Registry key	Value	When set
HKLM\SYSTEM\CurrentControlSet\Services \msdsm\Parameters \DsmMaximumRetryTimeDuringStateTransition	120	When MPIO support is specified and your server is Windows Server 2012 R2, Windows Server 2016 and Windows 2019, or Windows Server 2022
HKLM\SYSTEM\CurrentControlSet \Services\msdsm\Parameters \DsmMaximumStateTransitionTime	120	When MPIO support is specified and your server is Windows Server 2012 R2, or Windows Server 2016, Windows 2019 or Windows Server 2022
HKLM\SYSTEM\CurrentControlSet\Services \msdsm\Parameters\DsmSupportedDeviceList	"NETAPP LUN", "NETAPP LUN C-Mode" "NVMe NetApp ONTAO Con"	When MPIO support is specified
HKLM\SYSTEM\CurrentControlSet\Control \Class\ {iSCSI_driver_GUID}\ instance_ID \Parameters \IPSecConfigTimeout	60	Always
HKLM\SYSTEM\CurrentControlSet\Control \Class\ {iSCSI_driver_GUID}\ instance_ID \Parameters \LinkDownTime	10	Always
HKLM\SYSTEM\CurrentControlSet\Services \ClusDisk \Parameters\ManageDisksOnSystemBuses	1	Always

Registry key	Value	When set
HKLM\SYSTEM\CurrentControlSet\Control \Class\ {iSCSI_driver_GUID}\ instance_ID \Parameters \MaxRequestHoldTime	120	When no MPIO support is selected
HKLM\SYSTEM\CurrentControlSet\Control \Class\ {iSCSI_driver_GUID}\ instance_ID \Parameters \MaxRequestHoldTime	30	Always
HKLM\SYSTEM\CurrentControlSet\Control \MPDEV\MPIOSupportedDeviceList	"NETAPP LUN", "NETAPP LUN C-Mode", "NVMe NetApp ONTAO Con"	When MPIO support is specified
HKLM\SYSTEM\CurrentControlSet\Services\mpio \Parameters\PathRecoveryInterval	30	When your server is Windows Server 2012 R2, Windows Server 2016, Windows Server 2019, or Windows Server 2022
HKLM\SYSTEM\CurrentControlSet\Services\mpio \Parameters\PathVerifyEnabled	1	When MPIO support is specified
HKLM\SYSTEM\CurrentControlSet\Services \msdsm\Parameters\PathVerifyEnabled	1	When MPIO support is specified and your server is Windows Server 2012 R2, Windows Server 2016, Windows Server 2019, or Windows Server 2022
HKLM\SYSTEM\CurrentControlSet\Services \vnetapp\Parameters\PathVerifyEnabled	0	When MPIO support is specified
HKLM\SYSTEM\CurrentControlSet\Services \mpio\Parameters\PDORemovePeriod	130	When MPIO support is specified
HKLM\SYSTEM\CurrentControlSet\Services\msdsm \Parameters\PDORemovePeriod	130	When MPIO support is specified and your server is Windows Server 2012 R2, Windows Server 2016 , Windows Server 2019, or Windows Server 2022
HKLM\SYSTEM\CurrentControlSet\Services\vnetapp \Parameters\PDORemovePeriod	130	When MPIO support is specified, except if Data ONTAP DSM is detected

Registry key	Value	When set
HKLM\SYSTEM\CurrentControlSet\Services\mpio\Parameters\RetryCount	6	When MPIO support is specified
HKLM\SYSTEM\CurrentControlSet\Services\msdsm\Parameters\RetryCount	6	When MPIO support is specified and your server is Windows Server 2012 R2, Windows Server 2016, Windows Server 2019, or Windows Server 2022
HKLM\SYSTEM\CurrentControlSet\Services\mpio\Parameters\RetryInterval	1	When MPIO support is specified
HKLM\SYSTEM\CurrentControlSet\Services\msdsm\Parameters\RetryInterval	1	When MPIO support is specified and your server is Windows Server 2012 R2, Windows Server 2016, Windows Server 2019, or Windows Server 2022
HKLM\SYSTEM\CurrentControlSet\Services\vnetapp\Parameters\RetryInterval	1	When MPIO support is specified
HKLM\SYSTEM\CurrentControlSet\Services\disk\TimeOutValue	120	When no MPIO support is selected
HKLM\SYSTEM\CurrentControlSet\Services\mpio\Parameters\UseCustomPathRecoveryInterval	1	When MPIO support is specified and your server is Windows Server 2012 R2, Windows Server 2016, Windows Server 2019, or Windows Server 2022

NVMe parameters

The following NVMe Emulex driver parameters are updated when installing Windows Unified Host Utilities 7.2:

- EnableNVMe = 1
- NVMEMode = 0
- LimTransferSize=1

Registry values set by Windows Unified Host Utilities 7.1

The Windows Unified Host Utilities installer automatically sets registry values that are based on the choices that you make during installation. You should be aware of these registry values, the operating system version.

The following values are set by the Windows Unified Host Utilities installer. All values are in decimal unless otherwise noted.

> `HKLM` is the abbreviation for `HKEY_LOCAL_MACHINE`.

Registry key	Value	When set
HKLM\SYSTEM\CurrentControlSet\Services \msdsm\Parameters \DsmMaximumRetryTimeDuringStateTransition	120	When MPIO support is specified and your server is Windows Server 2008, Windows Server 2008 R2, Windows Server 2012, Windows Server 2012 R2, or Windows Server 2016, except if Data ONTAP DSM is detected
HKLM\SYSTEM\CurrentControlSet\Services \msdsm\Parameters \DsmMaximumStateTransitionTime	120	When MPIO support is specified and your server is Windows Server 2008, Windows Server 2008 R2, Windows Server 2012, Windows Server 2012 R2, or Windows Server 2016, except if Data ONTAP DSM is detected
HKLM\SYSTEM\CurrentControlSet\Services\msdsm \Parameters\DsmSupportedDeviceList	"NETAPP LUN"	When MPIO support is specified
	"NETAPP LUN", "NETAPP LUN C-Mode"	When MPIO support is specified, except if Data ONTAP DSM is detected
HKLM\SYSTEM\CurrentControlSet\Control\Class {iSCSI_driver_GUID}\ instance_ID\Parameters \IPSecConfigTimeout	60	Always, except when Data ONTAP DSM is detected
HKLM\SYSTEM\CurrentControlSet\Control \Class{iSCSI_driver_GUID} \ instance_ID\Parameters\LinkDownTime	10	Always
HKLM\SYSTEM\CurrentControlSet\Services\ClusDisk \Parameters\ManageDisksOnSystemBuses	1	Always, except when Data ONTAP DSM is detected
HKLM\SYSTEM\CurrentControlSet\Control \Class{iSCSI_driver_GUID} \ instance_ID\Parameters\MaxRequestHoldTime	120	When no MPIO support is selected
	30	Always, except when Data ONTAP DSM is detected
HKLM\SYSTEM\CurrentControlSet \Control\MPDEV\MPIOSupportedDeviceList	"NETAPP LUN"	When MPIO support is specified
	"NETAPP LUN", "NETAPP LUN C-Mode"	When MPIO is support-specified, except if Data ONTAP DSM is detected
HKLM\SYSTEM\CurrentControlSet\Services\mpio \Parameters\PathRecoveryInterval	40	When your server is Windows Server 2008, Windows Server 2008 R2, Windows Server 2012, Windows Server 2012 R2, or Windows Server 2016 only
HKLM\SYSTEM\CurrentControlSet\Services\mpio \Parameters\PathVerifyEnabled	0	When MPIO support is specified, except if Data ONTAP DSM is detected

Registry key	Value	When set
HKLM\SYSTEM\CurrentControlSet\Services\msdsm \Parameters\PathVerifyEnabled	0	When MPIO support is specified, except if Data ONTAP DSM is detected
HKLM\SYSTEM\CurrentControlSet\Services \msdsm\Parameters\PathVerifyEnabled	0	When MPIO support is specified and your server is Windows Server 2008, Windows Server 2008 R2, Windows Server 2012, Windows Server 2012 R2, or Windows Server 2016, except if Data ONTAP DSM is detected
HKLM\SYSTEM\CurrentControlSet\Services \msiscdsm\Parameters\PathVerifyEnabled	0	When MPIO support is specified and your server is Windows Server 2003, except if Data ONTAP DSM is detected
HKLM\SYSTEM\CurrentControlSet\Services\vnetapp \Parameters\PathVerifyEnabled	0	When MPIO support is specified, except if Data ONTAP DSM is detected
HKLM\SYSTEM\CurrentControlSet\Services\mpio \Parameters\PDORemovePeriod	130	When MPIO support is specified, except if Data ONTAP DSM is detected
HKLM\SYSTEM\CurrentControlSet\Services\msdsm \Parameters\PDORemovePeriod	130	When MPIO support is specified and your server is Windows Server 2008, Windows Server 2008 R2, Windows Server 2012, Windows Server 2012 R2, or Windows Server 2016, except if Data ONTAP DSM is detected
HKLM\SYSTEM\CurrentControlSet\Services\msiscdsm \Parameters\PDORemovePeriod	130	When MPIO support is specified and your server is Windows Server 2003, except if Data ONTAP DSM is detected
HKLM\SYSTEM\CurrentControlSet\Services \vnetapp \Parameters\PDORemovePeriod	130	When MPIO support is specified, except if Data ONTAP DSM is detected
HKLM\SYSTEM\CurrentControlSet\Services \mpio\Parameters\RetryCount	6	When MPIO support is specified, except if Data ONTAP DSM is detected
HKLM\SYSTEM\CurrentControlSet\Services\msdsm \Parameters\RetryCount	6	When MPIO support is specified and your server is Windows Server 2008, Windows Server 2008 R2, Windows Server 2012, Windows Server 2012 R2, or Windows Server 2016, except if Data ONTAP DSM is detected
HKLM\SYSTEM\CurrentControlSet\Services \msiscdsm\Parameters\RetryCount	6	When MPIO support is specified and your server is Windows Server 2003, except if Data ONTAP DSM is detected
HKLM\SYSTEM\CurrentControlSet\Services \vnetapp\Parameters\RetryCount	6	When MPIO support is specified, except if Data ONTAP DSM is detected
HKLM\SYSTEM\CurrentControlSet\Services \mpio\Parameters\RetryInterval	1	When MPIO support is specified, except if Data ONTAP DSM is detected

Registry key	Value	When set
HKLM\SYSTEM\CurrentControlSet\Services \msdsm\Parameters\RetryInterval	1	When MPIO support is specified and your server is Windows Server 2008, Windows Server 2008 R2, Windows Server 2012, Windows Server 2012 R2, or Windows Server 2016, except if Data ONTAP DSM is detected
HKLM\SYSTEM\CurrentControlSet\Services \vnetapp\Parameters\RetryInterval	1	When MPIO support is specified, except if Data ONTAP DSM is detected
HKLM\SYSTEM\CurrentControlSet \Services\disk\TimeOutValue	120	When no MPIO support is selected, except if Data ONTAP DSM is detected
	60	When MPIO support is specified, except if Data ONTAP DSM is detected
HKLM\SYSTEM\CurrentControlSet\Services\mpio \Parameters\UseCustomPathRecoveryInterval	1	When your server is Windows Server 2008,Windows Server 2008 R2, Windows Server 2012, Windows Server 2012 R2, or Windows Server 2016 only

See the Microsoft documents for the registry parameter details.

FC HBA values set by Windows Host Utilities

On systems using FC, the Host Utilities installer sets the required timeout values for Emulex and QLogic FC HBAs.

For Emulex FC HBAs, the installer sets the following parameters:

When MPIO is selected

Property type	Property value
LinkTimeOut	1
NodeTimeOut	10

When MPIO is not selected

Property type	Property value
LinkTimeOut	30
NodeTimeOut	120

For QLogic Fibre Channel HBAs, the installer sets the following parameters:

When MPIO is selected

Property type	Property value
LinkDownTimeOut	1
PortDownRetryCount	10

When MPIO is not selected

Property type	Property value
LinkDownTimeOut	30
PortDownRetryCount	120

> The names of the parameters might vary slightly depending on the program.
> For example, in the QLogic QConvergeConsole program, the parameter is displayed as `Link Down Timeout`.
> The Host Utilities `fcconfig.ini` file displays this parameter as either `LinkDownTimeOut` or `MpioLinkDownTimeOut`, depending on whether MPIO is specified. However, all of these names refer to the same HBA parameter. See Emulex or QLogic to learn more about the timeout parameters.

Understand the Host Utilities changes to FC HBA driver settings

During the installation of the required Emulex or QLogic HBA drivers on an FC system, several parameters are checked and, in some cases, modified.

The Host Utilities set values for the following parameters if MS DSM for Windows MPIO is detected:

- LinkTimeOut – defines the length of time in seconds that the host port waits before resuming I/O after a physical link is down.
- NodeTimeOut – defines the length of time in seconds before the host port recognizes that a connection to the target device is down.

When troubleshooting HBA issues, check to make sure these settings have the correct values. The correct values depend on two factors:

- The HBA vendor
- Whether you are using multipathing software (MPIO)

You can correct the HBA settings by running the Repair option of the Windows Host Utilities installer.

Emulex HBA drivers

If you have an FC system, you must verify the Emulex HBA driver settings. These settings must exist for each port on the HBA.

Steps

1. Open OnCommand Manager.
2. Select the appropriate HBA from the list and click the **Driver Parameters** tab.

   The driver parameters appear.

   a. If you are using MPIO software, ensure that you have the following driver settings:
      - LinkTimeOut - 1
      - NodeTimeOut - 10
   b. If you are not using MPIO software, ensure that you have the following driver settings:
      - LinkTimeOut - 30
      - NodeTimeOut - 120

QLogic HBA drivers

On FC systems, you must verify the QLogic HBA driver settings. These settings must exist for each port on the HBA.

Steps

1. Open QConvergeConsole, and then click **Connect** on the toolbar.

   The **Connect to Host** dialog box appears.

2. Select the appropriate host from the list, and then select **Connect**.

   A list of HBAs appears in the FC HBA pane.

3. Select the appropriate HBA port from the list, and then select the **Settings** tab.
4. Select **Advanced HBA Port Settings** from the **Select Settings** section.
5. If you are using MPIO software, verify that you have the following driver settings:
   - Link Down Timeout (linkdwnto) - 1
   - Port Down Retry Count (portdwnrc) - 10
6. If you are not using MPIO software, verify that you have the following driver settings:
   - Link Down Timeout (linkdwnto) - 30
   - Port Down Retry Count (portdwnrc) - 120

Troubleshoot

You can use the general troubleshooting techniques for Windows Host Utilities. Be sure to check the latest Release Notes for known issues and solutions.

The following is a list of the different areas you can investigate for potential interoperability issues:

- To identify potential interoperability issues, confirm that the Host Utilities support your combination of host operating system software, host hardware, ONTAP software, and storage system hardware. See the NetApp Interoperability Matrix Tool for more information.
- Verify that you have the correct iSCSI configuration.
- If iSCSI LUNs are not available after a reboot, verify that the target is listed as persistent on the **Persistent Targets** tab of the Microsoft iSCSI initiator GUI.
- If applications using the LUNs display errors on startup, verify that the applications are configured to depend on the iSCSI service.
- For FC paths to storage controllers running ONTAP, verify that the FC switches are zoned using the WWPNs of the target LIFs, not the WWPNs of the physical ports on the node.
- Review the Release Notes for Windows Host Utilities to check for known issues. The Release Notes include a list of known issues and limitations.
- Review the troubleshooting information in the SAN Administration Guide for your version of ONTAP.
- Search NetApp Bugs Online for recently discovered issues.
  - In the Bug Type field under Advanced Search, select **iSCSI - Windows** and then select **Go**. You should repeat the search for Bug Type **FCP -Windows**.
- Collect information about your system.
- Record any error messages that are displayed on the host or storage system console.
- Collect the host and storage system log files.
- Record the symptoms of the problem and any changes made to the host or storage system just before the problem appeared.
- If you are unable to resolve the problem, contact NetApp technical support for assistance.

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